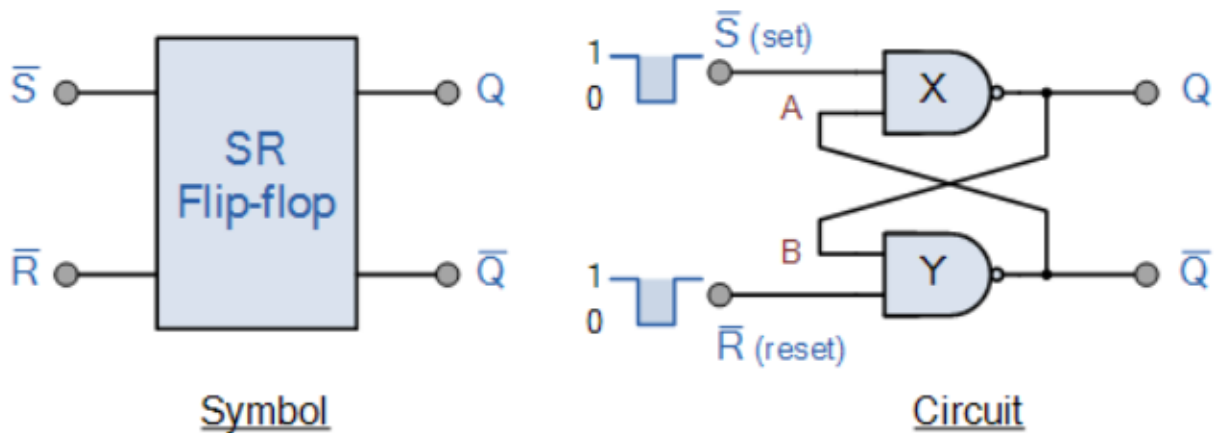


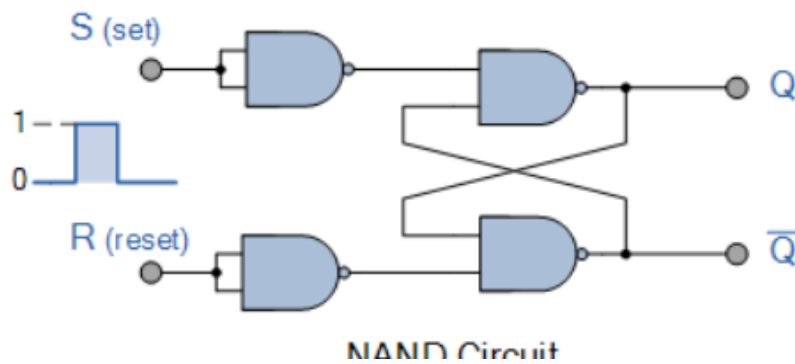
Circuit Simulation Lab:9

S-R Flip-Flop

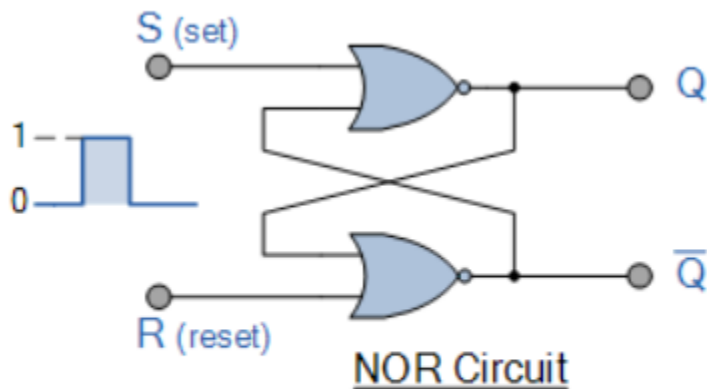
The purpose of this experiment is to introduce the design of simple sequential circuits, in this case S-R Flip Flop. The SR flip flop is a 1-bit memory bistable device having two inputs, i.e., SET and RESET. The SET input 'S' set the device or produce the output 1, and the RESET input 'R' reset the device or produce the output 0. The SET and RESET inputs are labeled as S and R, respectively. It is a single bit storage element. It has only two logic gates. The output of each gate is connected to the input of another gate



State	S	R	Q	$\bar{Q}$	Description
Set	1	0	0	1	Set $\bar{Q} \rightarrow 1$
	1	1	0	1	no change
Reset	0	1	1	0	Reset $\bar{Q} \rightarrow 0$
	1	1	1	0	no change
Invalid	0	0	1	1	Invalid Condition



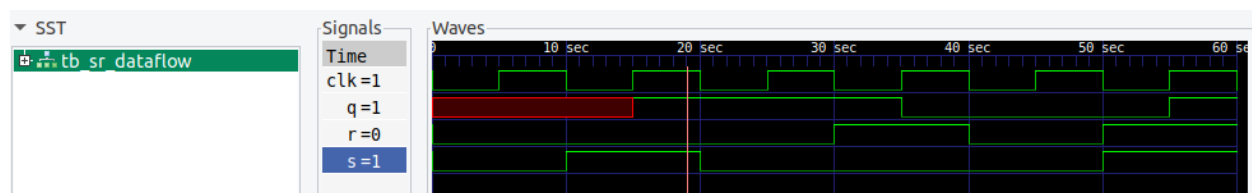
S	R	Q	$\bar{Q}$
0	0	No change	
0	1	0	1
1	0	1	0
1	1	X	X
(Invalid)			



S	R	Q	$\bar{Q}$
0	0	No change	
0	1	1	0
1	0	0	1
1	1	X	X
(Invalid)			

S-R Flip Flop

Dataflow Modeling
Code:



```

time=0 s=0 r=0 clk=0 q=x
time=5 s=0 r=0 clk=1 q=x
time=10 s=1 r=0 clk=0 q=x
time=15 s=1 r=0 clk=1 q=1
time=20 s=0 r=0 clk=0 q=1
time=25 s=0 r=0 clk=1 q=1
time=30 s=0 r=1 clk=0 q=1
time=35 s=0 r=1 clk=1 q=0
time=40 s=0 r=0 clk=0 q=0
time=45 s=0 r=0 clk=1 q=0
time=50 s=1 r=1 clk=0 q=0
time=55 s=1 r=1 clk=1 q=1
time=60 s=1 r=1 clk=0 q=1

```

Behavioral Modeling

Code:

```

module sr_behavioral(
input s,
input r,
input clk,
output reg q,
output reg qb
);

```

```

always @(posedge clk)
begin

```

```

case({s,r})

```

```

2'b00: begin

```

```

q <= q;
qb <= qb;
end

```

```

2'b01: begin

```

```

q <= 0;
qb <= 1;
end

```

```

2'b10: begin

```

```

q <= 1;
qb <= 0;
end

```

```

2'b11: begin
q <= 1'bx;
qb <= 1'bx;
end

endcase

end

Endmodule

Tb:
module tb_sr_behavioral;

reg s,r,clk;
wire q,qb;

sr_behavioral dut(
.s(s),
.r(r),
.clk(clk),
.q(q),
.qb(qb)
);

always #5 clk = ~clk;

initial begin

$dumpfile("sr_behavioral.vcd");
$dumpvars(0,tb_sr_behavioral);

clk=0;

s=0;r=0; #10;
s=1;r=0; #10;
s=0;r=0; #10;
s=0;r=1; #10;
s=0;r=0; #10;
s=1;r=1; #10;

$finish;

end

```

```
initial
$monitor("time=%0t s=%b r=%b clk=%b q=%b qb=%b",
        $time,s,r,clk,q,qb);
```

```
Endmodule
```

